

Question 1 (10 marks)**Marks**

- (a) The letters of the word CALCULUS are arranged in a row.
- (i) How many different arrangements are possible ? **1**
 - (ii) If one of these arrangements is selected at random, what is the probability that it begins with a “U” and ends with a “U” ? **2**
- (b) Liz heats a mug of milk up to 90°C in a microwave. She takes it out into a room where the temperature is a constant of 26°C . The milk cools to 70°C in 5 minutes. At time t minutes, its temperature T° decreases according to the equation
- $$\frac{dT}{dt} = -k(T - 26), \text{ where } k \text{ is a positive constant.}$$
- (i) Verify that $T = 26 + Ae^{-kt}$ is a solution of this equation, where A is a constant. **1**
 - (ii) Find the values of A and k . **2**
 - (iii) How long will it take for the temperature to cool to 30°C ? Give your answer to the nearest minute. **2**
 - (iv) Sketch the graph of T as a function of t . **2**

Question 2 (10 marks) Start a new page

- (a) A particle P moves along a straight line so that at time t , its displacement from a fixed point O on that line is given by
- $$x(t) = 3t^2(4 + t^3)^{-1}.$$
- (i) Find an expression for the velocity of the particle at time t . **1**
 - (ii) Find the time when the particle is momentarily at rest after the motion has started. **1**
 - (iii) Show that P is in exactly the same position at both times $t_1 = 1$ and $t_2 = 2 + 2\sqrt{2}$. **2**
 - (iv) Graph the displacement – time function. **2**
- (b) Two particles P and Q move along a line, their displacement at time t with respect to a fixed point O being $x(t)$ and $X(t)$ respectively.
- (i) The acceleration of P is given by $\frac{d^2x}{dt^2} = 6 + e^{-t}$. If it begins its motion at $x=0$ with a velocity of -1 , find an expression for $x(t)$. **2**
 - (ii) If $X(t) = 2 \sin 5t + 3t^2 + 2$, prove that $X(t) > x(t)$ for all $t \geq 0$. **2**

Question 3 (10 marks) Start a new page**Marks**

- (a) Prove that $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \frac{d^2x}{dt^2}$, where v is velocity and $\frac{d^2x}{dt^2}$ is acceleration as a function of time. **1**
- (b) The acceleration of a particle moving in a straight line is given by $\frac{d^2x}{dt^2} = (4x - 4)$, where x is the displacement, in metres, from the origin O and t is the time in seconds. Initially the particle is 6 metres to the right of O and its velocity (v m/s) is -8 m/s.
- (i) Show that $v^2 = 4x^2 - 8x - 32$. **2**
- (ii) Find the set of possible values of x where motion can exist and describe the motion of the particle. **3**
- (c) A particle is moving with simple harmonic motion in a straight line with a period of π seconds. Its maximum speed is 12 m/s. Initially the particle has a displacement of 3 metres from the centre of motion and is moving to the right.
- If x is the displacement, in metres, from the centre of motion ($x=0$) and t is the time in seconds, find an expression for x in terms of t . **4**

Question 4 (10 marks) Start a new page

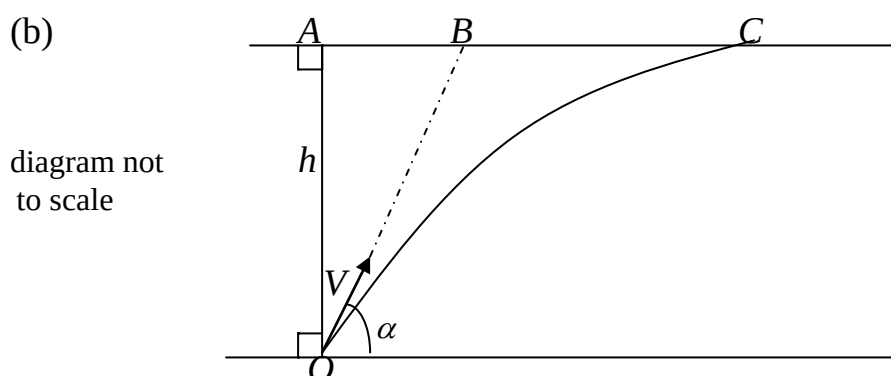
- (a) A particle is moving in a straight line. At time t seconds its displacement x metres from a fixed point O on the line is given by $x = 2 \cos^2 t$.
- (i) Prove that the motion is simple harmonic. **2**
- (ii) Find the amplitude of the motion. **1**
- (b) An employer wishes to choose two people for a job. There are eight applicants, three of whom are women and five of whom are men.
- (i) If each applicant is interviewed separately and all of the women are interviewed before any of the men, find how many ways there are in carrying out the interviews. **1**
- (ii) If the employer chooses two of the applicants at random, find the probability that at least one of those chosen is a woman. **2**

- (c) A nine-member Fund Raising Committee consists of four students, three teachers and two parents. The Committee meets around a circular table.
- (i) How many different arrangements of the nine members around the table are possible if the students sit together as a group, as do the teachers, but no teachers sit next to a student ? 2
- (ii) One student and one parent are related. Given that all arrangements in part (i) are equally likely, what is the probability that these two members sit next to each other ? 2

Question 5 (10 marks)

Start a new page

- (a) Find how many groups of one or more digits can be formed from the following digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 if repetition is not allowed. 2



In the diagram, an aeroplane is flying with constant velocity U at a constant height h above horizontal ground.

When the plane is at A , it is directly over a gun at O .

When the plane is at B (time $t = 0$), a shell is fired from the gun at the plane along the direction OB . The shell is fired with initial velocity V at an angle of elevation α .

The horizontal and vertical components of the displacement of the shell from O at time t are given respectively by

$$x = Vt \cos \alpha \quad \text{and} \quad y = Vt \sin \alpha - \frac{1}{2}gt^2,$$

while g is the acceleration due to gravity.

- (i) Show that if the shell hits the plane at point C at time $t = T$, then 2

$$VT \cos \alpha = \frac{h}{\tan \alpha} + UT.$$

- (ii) Show that when the shell hits the plane then $2U(V \cos \alpha - U) \tan^2 \alpha = gh$. 3

Marks

- (c) The velocity v of a particle at time t , is given in terms of its displacement, x , by the equation $v = \frac{4}{x}$, where $x \neq 0$. Initially, $x = 8$.
- (i) Find an expression for the acceleration of the particle in terms of x . **1**
- (ii) By expressing v as $\frac{dx}{dt}$, find an expression for x^2 in terms of t . **2**

Question 6 (10 marks)**Start a new page**

- (a) Prove that ${}^{n+1}C_{k+1} = {}^nC_k + {}^nC_{k+1}$, for $1 \leq k < n$ and $n \geq 1$. **2**
(Do not use induction)
- (b) The rate of change of the population of a country is affected by the maximum possible population M of the country. M depends on factors such as the area of land and the amount of raw materials etc.
If P is the population it can be shown that $\frac{dP}{dt} = kP(M - P)$,
where M and k are constants and t is measured in years,
- (i) Verify that $P = \frac{A M e^{Mkt}}{A e^{Mkt} + 1}$ is a solution to the equation where A is a constant. **2**
- (ii) It is known that the maximum possible population (M) of a country is 860 million. In 1790 the population of the country was 4 million people and in 1800 the population was 6 million people. **5**
In what year was the population of the country equal to half of its maximum possible population (i.e. 430 million)? Give your answer to the nearest year.
- (iii) Describe what happens to the population growth rate as P approaches M . **1**

END of PAPER

Solutions to 2005 T2 Y12 Ext 1.

Q1 a) 2C 1A 2u 2L 1S Total 8 letters

$$\frac{8!}{2!2!2!} = \underline{\underline{5040}}$$

ii) No of arrangements begins with U and ends in U
 $= \frac{6!}{2!2!} = 180$

$$\text{Prob} = \frac{180}{5040} = \underline{\underline{\frac{1}{28}}}$$

b) i) $T = 26 + Ae^{-kt}$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T - 26)e^{-kt} \quad (\text{since } A = T - 26)$$

ii) $t=0, T=90^\circ$

$$90 = 26 + A$$

$$\underline{A = 64}$$

$$\therefore T = 26 + 64e^{-kt}$$

when $t=5, T=70^\circ$

$$70 = 26 + 64e^{-5k}$$

$$\frac{44}{64} = e^{-5k}$$

$$-5k = \ln\left(\frac{44}{64}\right)$$

$$\underline{k = 0.07494 \text{ (4 sf)}}$$

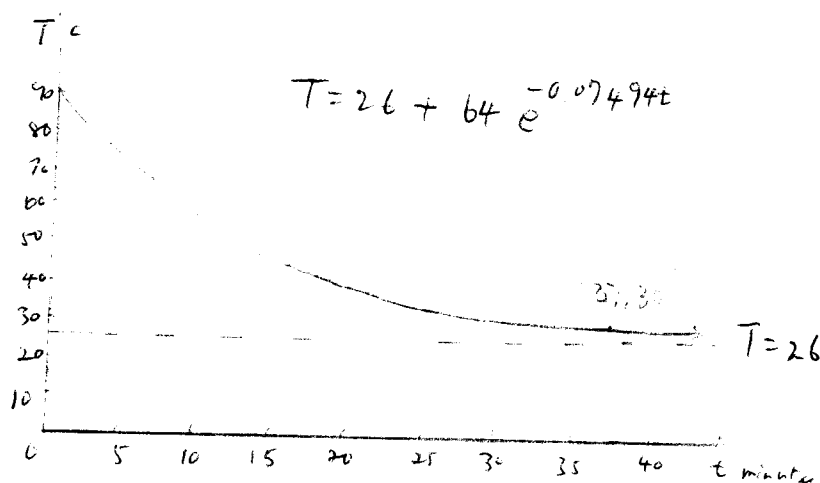
iii) $30 = 26 + 64e^{-0.07494t}$

$$\frac{4}{64} = e^{-0.07494t}$$

$$\ln\left(\frac{1}{16}\right) = (-0.07494)t$$

$$\underline{t = 37 \text{ min (nearest min)}}$$

ibiv



Q 2

a) $x = \frac{3t^2}{4+t^3}$

i) $V = \frac{(4+t^3)6t - 3t^2(3t^2)}{(4+t^3)^2}$

$$= \frac{24t + 6t^4 - 9t^4}{(4+t^3)^2}$$

$$= \frac{24t - 3t^4}{(4+t^3)^2}$$

ii) $V=0$ when $24t - 3t^4 = 0$
 $8t - t^4 = 0$
 $t(8 - t^3) = 0$
 $t=0$ or $t=2$
 $t=2$

iii) $t_1 = 1, x_1 = \frac{3}{4+1} = \frac{3}{5}$

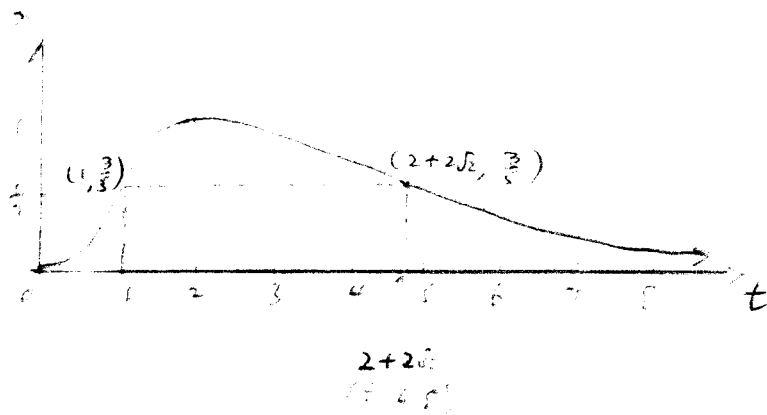
$t_2 = 2+2\sqrt{2}, x_2 = \frac{3(2+2\sqrt{2})^2}{4+(2+2\sqrt{2})^3}$

$$\begin{aligned} x_2 &= \frac{3(4 + 8\sqrt{2} + 8)}{4 + 2^3 + 3(2^2 \cdot 2\sqrt{2}) + 3(2 \times 4 \cdot 2) + 8 \cdot 2\sqrt{2}} \\ &= \frac{3(12 + 8\sqrt{2})}{12 + 24\sqrt{2} + 48 + 16\sqrt{2}} \\ &= \frac{36 + 24\sqrt{2}}{60 + 40\sqrt{2}} \end{aligned}$$

$$\text{At } t_2 \quad x_2 = \frac{12(3+2\sqrt{2})}{20(3+2\sqrt{2})} = \frac{3}{5}$$

$$\therefore x_1 = x_2$$

iv)



$$\begin{aligned} 2b) \quad \ddot{x} &= 6 + e^{-t} & \bar{x} &= \int 6 + e^{-t} dt \\ \dot{x} &= 6t - e^{-t} + k_1 \\ t=0, \quad \dot{x} &= -1 & \therefore -1 &= -e^{-t} + k_1 \\ & & \therefore k_1 &= 0 \end{aligned}$$

$$\therefore \dot{x} = 6t - e^{-t} \quad |$$

$$x = \int 6t - e^{-t} dt$$

$$x = 3t^2 + e^{-t} + k_2$$

$$\begin{aligned} t=0, \quad x &= 0, & 0 &= 1 + k_2 \\ & & -1 &= k_2 \end{aligned}$$

$$\therefore \underline{\underline{x = 3t^2 + e^{-t} - 1}} \quad |$$

$$\begin{aligned} \text{ii) } X(t) - x(t) &= 2 \sin 5t + \cancel{3t^2} + 2 - (\cancel{3t^2} + e^{-t} - 1) \\ &= 3 + 2 \sin 5t - e^{-t} \end{aligned}$$

$$\left. \begin{aligned} \min \sin 5t &= -1 \\ \min 2 \sin 5t &= -2 \end{aligned} \right\} 3 + 2 \sin 5t \geq 1$$

$\max e^{-t} = 1$ for $t \geq 0$ because e^{-t} is a decreasing function
and $e^0 = 1$ and $e^{-t} < 1$ for $t > 0$

$$\text{At } t=0 \quad X(0) - x(0) = 3 + 0 - 1 = 2 \quad \frac{1}{2}$$

$$\therefore X(t) - x(t) > 0 \quad \text{for } t \geq 0$$

Q 3

$$\begin{aligned}
 a) \quad \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \frac{dv}{dx} \\
 &= v \frac{dv}{dx} \\
 &= \frac{dx}{dt} \cdot \frac{dv}{dx} \\
 &= \frac{dv}{dt} \\
 &= \frac{d^2 x}{dt^2} \quad \#
 \end{aligned}$$

$$b) \quad \frac{d(\frac{1}{2} v^2)}{dx} = 4x - 4$$

$$v^2 = 2 \int (4x - 4) dx$$

$$v^2 = 4x^2 - 8x + K$$

$$t=0, \quad x=6, \quad v=-8$$

$$64 = 4(36) - 48 + K$$

$$-32 = K$$

$$\therefore v^2 = 4x^2 - 8x - 32$$

$$ii) \quad v^2 = 4(x-4)(x+2) \geq 0$$

$$\therefore \underline{x \geq 4 \text{ or } x \leq -2} \text{ for motion to exist}$$

$$\text{In this case initially } x=6 \quad v=-8 \quad \ddot{x}=20$$

The particle starts at 6m to the right of O, moving to the left, slowing down ($\ddot{x}=20 > 0$) until it reaches 4m to the right of O. It stops there momentarily, turns around and moves to the right, speeding up forever and never returns.

3c) Period = π

$$\frac{2\pi}{h} = 2 \Rightarrow n = 2$$

max $|v| = 12$ at $x = 0$

$$v^2 = h^2(a^2 - x^2)$$

$$144 = 4(a^2 - 0)$$

$$a = 6$$

$t = 0, x = 3$ moving to the right

$$x = 6 \sin(2t + \alpha)$$

$$3 = 6 \sin(\alpha)$$

$$\frac{1}{2} = \sin \alpha$$

$$\alpha = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

for rejecting $\frac{5\pi}{6}$

But $\dot{x} = 12 \cos(2t + \alpha)$

At $t = 0, \dot{x} = 12 \cos \frac{\pi}{6} = 6$ (moving to the right)

$\dot{x} = 12 \cos \frac{5\pi}{6} = -6$ (moving to the left)

$$\therefore \alpha = \frac{\pi}{6}$$

$$\therefore x = \underline{\underline{6 \sin(2t + \frac{\pi}{6})}}$$

Note: there are other alternative solutions.

Q 4 ai) $x = 2 \cos^2 t = 1 + \cos 2t$
 $x - 1 = \cos 2t$

$$\frac{dx}{dt} = -2 \sin 2t$$

$$\frac{d^2x}{dt^2} = -4 \cos 2t$$

$$= -2^2(x - 1)$$

$\therefore$ SHM

which is in the form of $\ddot{x} = -h^2(x - b)$

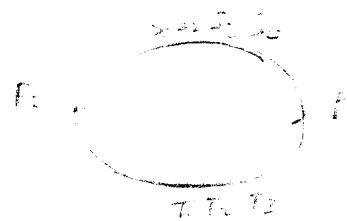
ii) amplitude = 1 m

Q 4 b) $3! 5! = 720$

∴ $1 - P(\text{All men}) = 1 - \frac{5}{8} \times \frac{4}{7} = \frac{9}{14}$

c) i) Parents 2 way
Students as a group $4!$
Teachers as a group $3!$

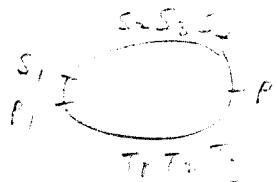
Total no. of ways = $2 \times 4! \times 3! = 288$



ii) S_1, P_1 can swap with P

Total no. of ways = $2 \times 3! \times 3! = 72$

Prob = $\frac{72}{288} = \frac{1}{4}$



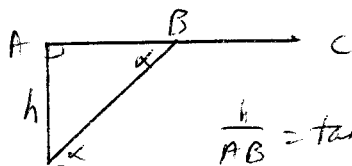
$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$

5 a) 10 different digits. Each can be in or out (2 choices)
but at least 1 digit needs to be included (ie exclude all digits out)

$2^{10} - 1 = \underline{\underline{1023}}$

b) $x = VT \cos \alpha$ horizontal distance travelled by shell from A to C
Horizontal distance travelled by plane from A to C

$= AB + BC$
 $= \frac{h}{\tan \alpha} + uT$



$\frac{h}{AB} = \tan \alpha$
 $AB = \frac{h}{\tan \alpha}$

From B C
Speed is u , constant
time is T

$\therefore VT \cos \alpha = \frac{h}{\tan \alpha} + uT$

5 biii) $V \tan \alpha = \frac{h}{\tan \alpha} + uT$ (from part ii)

Solve for T : $V \tan \alpha - uT = \frac{h}{\tan \alpha}$

$$T(V \tan \alpha - u) = \frac{h}{\tan \alpha}$$

$$T = \frac{h}{\tan \alpha (V \tan \alpha - u)}$$

$y = Vt \sin \alpha - \frac{1}{2}gt^2$ (from part i)

At C, $t = T$, $y = h$

$$h = VT \sin \alpha - \frac{1}{2}gT^2$$

$$h = V \frac{h}{\tan \alpha (V \tan \alpha - u)} \sin \alpha - \frac{1}{2}g \frac{h^2}{(\tan \alpha)^2 (V \tan \alpha - u)^2}$$

$$1 = \frac{V \sin \alpha}{(V \tan \alpha - u)} - \frac{1}{2} \cdot \frac{gh}{(\tan \alpha)^2 (V \tan \alpha - u)^2}$$

$$1 = \frac{2(V \tan \alpha)(V \tan \alpha - u)(\tan \alpha)^2 - gh}{2(\tan \alpha)^2 (V \tan \alpha - u)^2}$$

$$2(\tan \alpha)^2 (V \tan \alpha - u)^2 = 2V \tan \alpha (V \tan \alpha - u)(\tan \alpha)^2 - gh$$

$$2(V \tan \alpha - u)(\tan \alpha)^2 [V \tan \alpha - u - V \tan \alpha] = -gh$$

$$-2u(V \tan \alpha - u)(\tan \alpha)^2 = -gh$$

$$\underline{2u(V \tan \alpha - u)(\tan \alpha)^2 = gh} \quad \text{Q.E.D.}$$

$$5c) i) \ddot{x} = \frac{d(\dot{x})}{dx} = \frac{d\left(\frac{1}{2} \times \left(\frac{4}{x}\right)^2\right)}{dx} = \frac{d\left(\frac{8}{x^2}\right)}{dx} = \frac{-16}{x^3}$$

$$ii) \frac{dx}{dt} = \frac{4}{x}$$

$$\int x dx = \int 4 dt$$

$$\frac{x^2}{2} = 4t + k$$

$$t=0, x=8 \quad \frac{64}{2} = k \quad \dots \quad k=32$$

$$\therefore x^2 = 2(4t + 32)$$

$$\underline{\underline{x^2 = 8t + 64}}$$

$$\begin{aligned} 6a) \quad {}^nC_k + {}^nC_{k+1} &= \frac{n!}{(n-k)!k!} + \frac{n!}{(n-k-1)!(k+1)!} \\ &= \frac{n!(\cancel{k+1} + n - \cancel{k})}{(n-k)! (k+1)!} \\ &= \frac{(n+1)n!}{(n-k)!(k+1)!} \\ &= \frac{n!}{(n-k)!(k+1)!} \\ &= {}^{n+1}C_{k+1} \end{aligned}$$

$$b) \quad p = \frac{A M e^{Mkt}}{A e^{Mkt} + 1}$$

$$\frac{dp}{dt} = \frac{(A e^{Mkt} + 1) A M^2 k e^{Mkt} - A M e^{Mkt} (A M k e^{Mkt})}{(A e^{Mkt} + 1)^2}$$

$$\frac{dp}{dt} = \frac{A M^2 k e^{Mkt} + A M^2 k e^{Mkt} - A^2 M^2 e^{2Mkt}}{(A e^{Mkt} + 1)^2} = \frac{A M^2 k e^{Mkt}}{(A e^{Mkt} + 1)^2}$$

$$kP(M-P) = \frac{k \cdot A M e^{Mkt}}{A e^{Mkt} + 1} \left(M - \frac{A M e^{Mkt}}{A e^{Mkt} + 1} \right)$$

$$= \frac{k A M e^{Mkt}}{A e^{Mkt} + 1} \left(\frac{A \cancel{M e^{Mkt}} + M - A \cancel{M e^{Mkt}}}{A e^{Mkt} + 1} \right)$$

$$kP(M-P) = \frac{k A M e^{Mkt}}{(A e^{Mkt} + 1)^2}$$

$$\therefore \frac{dP}{dt} = kP(M-P) \quad \text{Q.E.D}$$

$$ii) \quad M = 860 \times 10^6 \quad \rightarrow \quad P = \frac{A \times 860 \times 10^6 \times e^{860 \times 10^6 k t}}{A e^{860 \times 10^6 k t} + 1}$$

$$(Y. 1790) \quad t=0 \quad P = 4 \times 10^6 = \frac{860 \times 10^6 A}{A + 1}$$

$$4(A+1) = 860 A$$

$$4 = 856 A$$

$$\therefore A = \frac{1}{214}$$

$$(Y. 1800) \quad t=10 \quad P = 6 \times 10^6 = \frac{\frac{1}{214} \times 860 \times 10^6 \times e^{860 \times 10^6 \times 10k}}{\frac{1}{214} e^{860 \times 10^6 \times 10k} + 1}$$

$$\frac{6}{214} e^{860 \times 10^6 \times 10k} + 6 = \frac{1}{214} \times 860 \times e^{860 \times 10^6 \times 10k}$$

$$e^{860 \times 10^6 \times 10k} \cdot \frac{856}{214} = 6$$

$$e^{860 \times 10^6 \times 10k} = \frac{6 \times 214}{856} = \frac{3}{2}$$

$$860 \times 10^6 \times 10k = \ln\left(\frac{3}{2}\right)$$

$$k = \left(\ln \frac{3}{2}\right) \div (860 \times 10^6 \times 10)$$

$$k = 4.7419 \times 10^{-11} \text{ (55.f)}$$

$$\text{Half of } M = 430$$

$$430 \times 10^6 = \frac{\frac{1}{214} \times 860 \times 10^6 \times e^{860 \times 10^6 \times 4.7419 \times 10^{-11} \times t}}{\frac{1}{214} e^{860 \times 10^6 \times 4.7419 \times 10^{-11} \times t} + 1}$$

$$\frac{1}{214} e^{860 \times 4.7419 \times 10^{-5} t} + 1 = \frac{2}{214} e^{860 \times 10^{-5} \times 4.7419 t}$$

$$1 = \frac{1}{214} e^{860 \times 10^{-5} \times 4.7419 t}$$

$$\ln(214) = (860 \times 10^{-5} \times 4.7419) t$$

$$t = \frac{\ln(214)}{860 \times 10^{-5} \times 4.7419}$$

$$t \doteq 132$$

$$\text{i.e. } 1790 + 132 = 1922$$

$$\underline{\underline{Yr 1922}}$$

$$\text{(ii) Since } \frac{dP}{dt} = kP(M-P), \text{ as } P \rightarrow M, \frac{dP}{dt} \rightarrow 0$$